

FUNCTION

In general, this paper provide a good way to represent function intuitively, graphically and algebraically.

Point – Slope Function

In modern mathematics, function is represent by relationship of one point to another point as well as line, curve, and etc. Line built up by one point and slope. Every point at line have same slope and it can reach by:

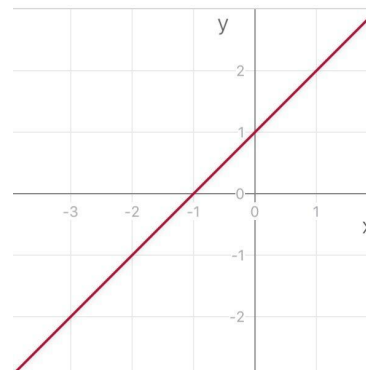
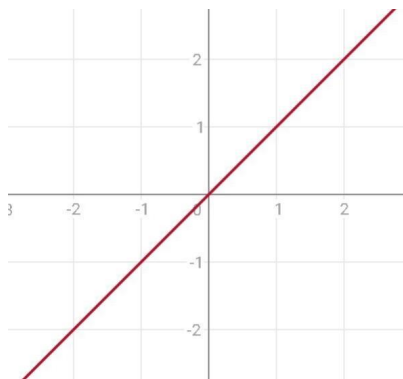
$$\text{slope} = m = \frac{\text{rise}}{\text{run}} = \frac{\Delta Y}{\Delta X}$$

$$\text{Tangent equation} = y - y_1 = m(x - x_1) = m(x - x_1) + y_1 = m \times a + y_1$$

Basicly, the equation derivative from slope by fix certain point of (x_n, y_n) of the graph. Remember that :

$$\Delta y = y_2 - y_1$$

$$\Delta x = x_2 - x_1$$



This graph represent humble function in math and it's transformation which there's a lot of application in calculus.

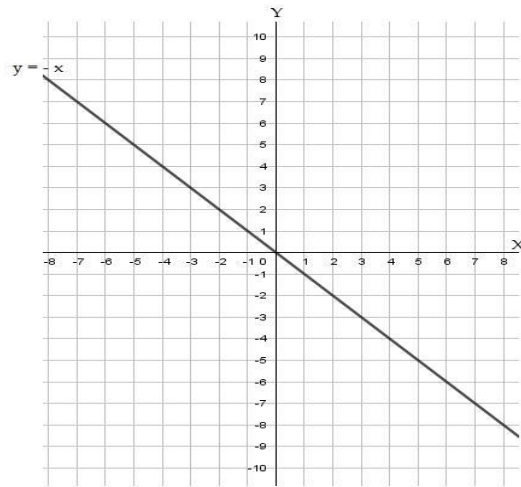
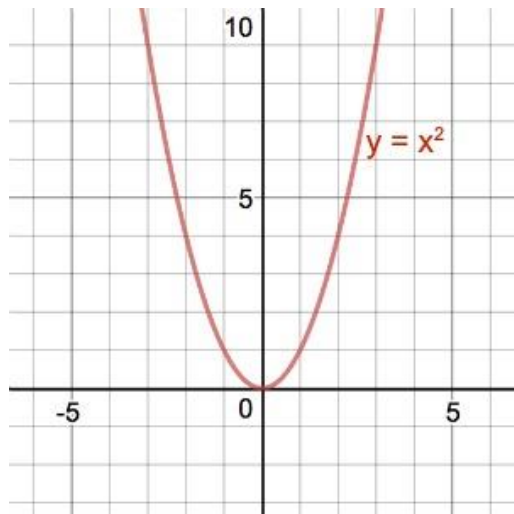
$$f(x) = x,$$

$$f(x) = x - 1$$

Notice that the constant (C) equals zero which is why lies in origin point. The right function correctly same with the left but have been transform by translation.

Then, function represent relationship between domain and range (input and output) which every input give one output. This represent that every algebra term statisfies for a function if input not give probability for the output meaning every function map every input to and product just one output. In general, there is unit function which every input

give specifically unique output. It means there is no way same input give difference value of y as well as both graph.



Notice in left graph every two input give one exactly output. This function can also represent the behavior of reflection to the y axis. Remember that:

$$\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -x \\ y \end{bmatrix}$$

$$f(-x) = f(x) = y$$

Then we can conclude that every function which can represent by reflection of y axis also represent an even function. Meaning every opposite input with the same direction to the mirror reach exactly one value in set of range. This is difference with right function because it represent one-to-one function and it's odd function. Notice that the left function can also represent as curve with secant line in every point between (x and -x) and slope equals zero. Then, every function can represent as unit function if there zero straight line on the graph except the x- axis (y=0).

In general calculus is all about exponential, logarithms, trigonometry and piecewise function with continuity problem (domain problem) :

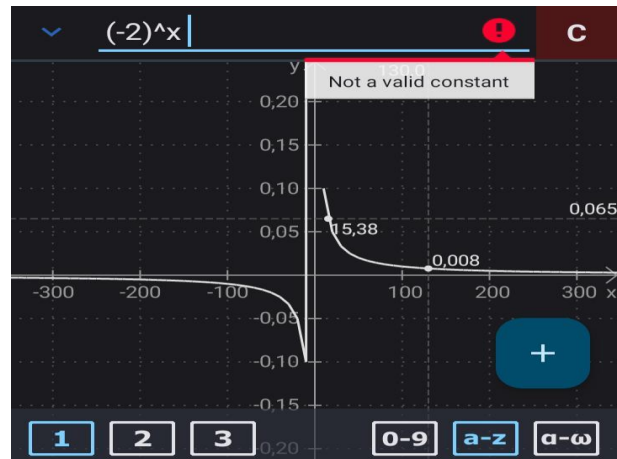
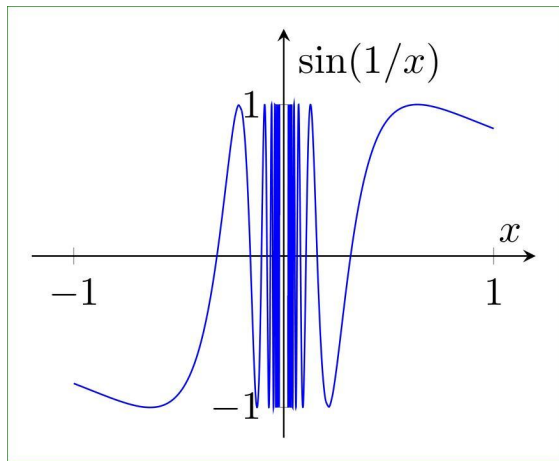
Exponential function

General term:

$$f(x) = b \times a^x + C$$

a = positif Integer, $x \geq 0$, b and c represent all real number which b is scalar value for multiplication of $f(x)$. C represent the y interception at $x=0$ or a lot of people known as value that up or down function.

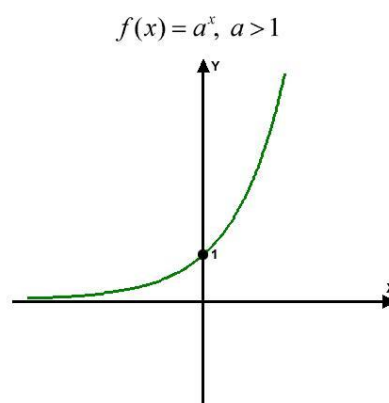
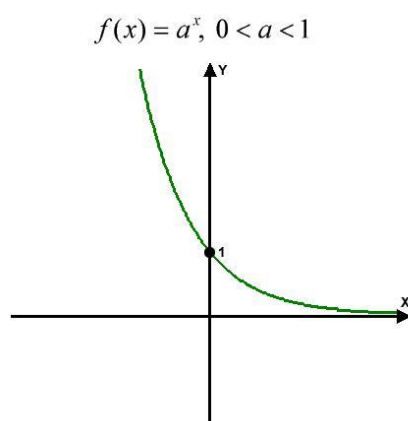
There reason why u cannot equals negative value because if it does then the output would up and down dependent on the number of power (up if it even and down if it odd). The problem is when the set of input (domain) reach large value such us (100, 1000, 10.000, 10^n). Intutively, we can imagine that the graph Will ocilated rapidly. In calculus this term Will be a problem because this behavior of function makes the function getting harder to determine the limit (limit does not exist) as well as:



The other reason, i think this might be the main reason is if x statisfies for negative value it give problem in real number such us:

Example : $f(x) = a^x$ for $a = -2$ and $x = \frac{1}{n}$ then $(-2)^{\frac{1}{n}}$ give an imaginary number

Look at both function below:



Why it look like that ? Remember the inverse axiom represent as

$$a \times a^{-1} = a \times \frac{1}{a} = 1 \text{ then } \frac{1}{a^{-1}} = a$$

The behavior of left function caused by algebraically to get y as an output we use 1/input which the value get bigger and bigger as x to $-\infty$.

Logarithms function

If : $a^x = b$ then $\log a b = x$

Did you notice the big picture of relationship of exponential and logarithms function ? It was the inverse function which means both function categories into one-to-one function. The reason why is because exponential function transform an input to output but logarithms function use output as an input to reach input as an output. I hope you not confuse because it's the big picture of both relationship.

Remember the $f(x)=x$ i say before ? this function represent as linear transformation of origin point of Algebra function. The reason i notice this is because the output function that reach by this transformation was an inverse function

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \times \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} y \\ x \end{bmatrix}$$

This matrix multiplication correctly answer the algebraically proof that :

$$f(x) = y, f(y) = x$$

Notice that $y=f(x)$ and $x=f(y)$ then we ca substitute as well as:

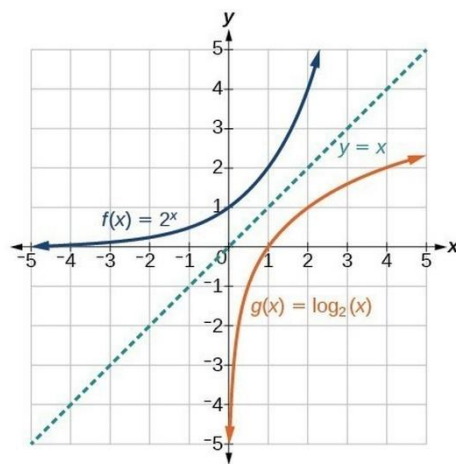
$$f(f(y)) = x \text{ and } f(f(x)) = y$$

This Equation represent function that treat y as domain have been substitute in another function reach x as codomain and a function which treat x as an input have been substitute to another function who can reach y as codomain or output. This definision is definision of inverse function such that :

$$f(f(x)^{-1}) = f(f(x))^{-1} = x$$

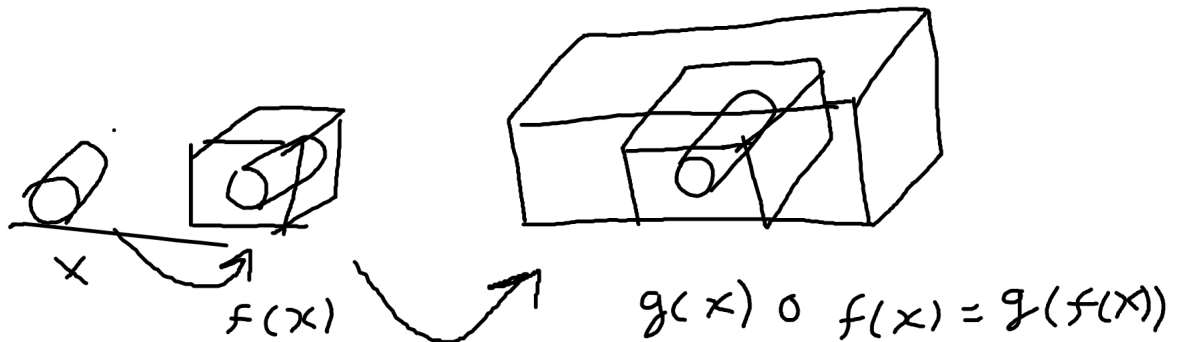
Examples : $f(x) = x^2$ then the inverse is $x^{\frac{1}{2}}$ because it stasfies for $\left(x^{\frac{1}{2}}\right)^2 = (x^2)^{\frac{1}{2}} = x$

Graphically the relationship between both function represent as :



Remember that $a^0 = 1$ caused $\log_a 1 = 0$ as the intersection for both of y axis and x axis

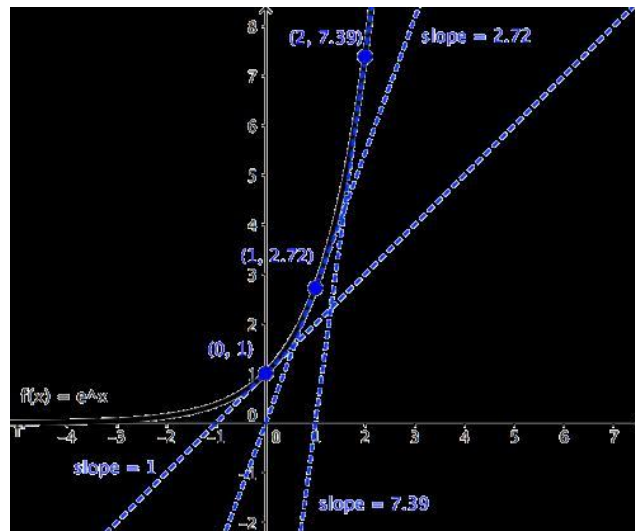
General Idea of Composition of Function :



Euler Constant

We Will learn more about this special constant in next paper, this is special because it represent irrational number as well as π , and etc which there is a lot of application in real world. The important thing that you need to know, this constant statisfies of non negative number to represent as exponential function. This function through at point (0,1) but the special is the derivative of that function at (0,1) equals perfectly one which means the slope of tangent line cross at point (0,1) of that curve is one. This impact as the derivative function and integral form remain same.

$$f(x) = e^x$$



Piecewise function

Basically, the piecewise function exist to handle the problem of function it self which every input cannot have more than one output using the tools of hole and full circle which represent not equals it and equals it. Graphically, every difference domain give difference output and behavior or rules such us increase to decrease or line to curve and so on

I hope you enjoy this paper as well i enjoy Mathematics much. I Will post other paper include to some problem in next time. Hope you enjoy and have a good day!

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